

# JEFF GE

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## PROFESSIONAL SUMMARY

- **AI Architect & Engineer** with 3+ years of industry experience driving AI transformation through strategic integration of LLMs, Graph Analytic, and NLP architectures.
- **Cross-Industry Impact:** Trusted to design and deliver highly scalable AI solutions for top-tier clients across Telecommunications (HKBN), Finance (Deloitte, HKMA), and Public Sectors (HKPF, AA).
- **Innovation to ROI:** Adept at translating cutting-edge AI into measurable business value. Expert in architecting RAG/Agent frameworks and driving enterprise adoption through client workshops on next-gen AI tools.

## WORKING EXPERIENCE

### Hong Kong Broadband Network

11/2025 - 08/2026, *Specialist, AI Application*

- **TenderAI:** Spearheaded the design and deployment of a GraphRAG-powered enterprise platform utilizing LlamaIndex, Neo4j, and PostgreSQL to optimize company tender process and drive AI-based analysis.
- **Hybrid Retrieval Pipeline Development:** Engineered a dynamic multi-strategy retrieval engine integrating vector semantic search, graph traversal, and keyword matching. Implemented a multi-file evidence extractor to ensure high-accuracy responses with precise page-level citations. Achieved a 95%+ retrieval accuracy and reduced query latency by 40% through optimized vector and graph traversals.
- **Technical Consulting & Pre-sales:** Partnered with sales teams to drive client engagements, translating business requirements into localized deployment strategies via comprehensive market solution sourcing. Liaised with external vendors for product evaluation, while delivering strategic technical advisory to senior leadership to guide decision-making.

### Hong Kong Applied Science and Technology Research Institute

03/2023 - 11/2025, *AI Engineer*

- **Neo4j Graph Algorithm Development:** Deployed community detection and node similarity algorithms on HKMA network analysis program. Constructed graph data model to map complex financial entity relationships. Delivered interactive visualizations of network to support regulatory risk assessment, enhancing proactive identification of suspicious networks linked to money laundering, market manipulation and fraud transactions. Mapped over 1M+ financial entities and reduced the time for risk identification from days to hours.
- **ChatDoc LLM Development:** Focused on the Hong Kong Police Force and Airport Authority program. Implemented a conversational RAG compatible with LLM. Developed and integrated a streamlined user feedback module into the ChatDoc back-end to capture and analyze user interactions.
- **ESG Initiatives NER Extraction:** Trained UIE models for ESG initiative recognition. Developed API enabling Deloitte users to automatically extract targets from annual financial reports. Provided back-end support for ESG reasoning systems.
- **AI-Powered Advertising Visualization:** Explored AI image generation using Stable Diffusion, with ComfyUI for workflow optimization and Control-Net for visual control. Developed end-to-end pipelines to train task-specific models for automated generation of custom background image to replace clients' reliance on Shutter-stock assets.

### Bank of China Financial Technology Co., Ltd., Shanghai

06/2022-09/2022, *Intelligent Risk Control*

- **Intelligent Property Valuation:** Investigated property right evaluation companies and available public data sources.
- **Trained Models:** Divided intellectual property rights to be evaluated into high, middle and low net worth for training.

## EDUCATION

### The Hong Kong Polytechnic University

09/2021 - 03/2023, *M.Sc. in Data Science and Analytics*

- **Related Courses:** *Optimization Methods, Principles of Data Science, Statistical Data Mining, Big Data Computing, Advanced High Dimensional Data Analysis, Deep Learning.*

### Hohai University, Nanjing, China

09/2017 - 06/2021, *B.Sc. in Information and Computing Science*

- **Related Courses:** *C++, Data Structure, Machine Learning, Stochastic Process, Multivariate Statistics, Probability, Mathematical Modeling, PDE, ODE, Numerical Analysis, Optimization Algorithm.*

## RESEARCH AND PROJECT EXPERIENCE

### HUAWEI CTF Security Competition

07/2022-08/2022, *Core Team Member*

- Constructed elliptic curve Abel group in discrete case over finite field based on modular arithmetic. Reversely factorized two big prime number  $p$  and  $q$  in  $RSA$ . Calculated the Euler function and decrypt the original key  $d$  to generate  $FLAG$ .

### Credit Suisse: CodeIT Competition

09/2022-10/2022, *Core Team Member*

- Participated in Programming Hackathon, cooperated with other partners on GitHub and eventually gained 7th of 24 teams.

## SKILLS

- **Programming:** Python (PyTorch, OpenCV, Pandas, NumPy, Ollama), Java, Linux, Git
- **Vibe Coding:** Claude Code, OpenClaw, Hermes Agent, Cursor, VSCode, CodeX
- **Database & Deployment:** Neo4j, PostgreSQL, Milvus, Docker, Kubernetes
- **LLMs & Machine Learning:** ComfyUI, Stable Diffusion, RAG, LightGBM, LoRA